REAL ESTATE PRICE PREDICTION BASED ON LOCATION AND INFRASTRUCTURE

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1 Introduction

In the context of rapid urbanization, real estate prices are strongly influenced by geographical location and surrounding amenities. Accurate price prediction plays an important role in supporting decision-making in investment and urban planning. Previous studies often assume a linear relationship between features such as distance to MRT stations and housing prices. However, the actual impact of nonlinear relationships and multicollinearity may vary depending on the dataset. Therefore, it is necessary to empirically evaluate their influence on model performance. In this study, we aim to compare linear models (OLS, Ridge, Lasso) and a nonlinear model (Support Vector Regression - SVR) for real estate price prediction. We further investigate the impact of multicollinearity and evaluate model performance under different data splitting strategies. A standard machine learning pipeline with data normalization and K-fold cross-validation is employed, and performance is evaluated using MAE, RMSE, and R^2 metrics.

2 Related Work

Real estate price prediction has been widely studied using various machine learning approaches.

For example, [1] applied linear regression models such as OLS, Ridge, and Lasso, showing that regularization techniques can improve model stability in the presence of correlated features. However, their study mainly focuses on linear relationships and does not explore nonlinear modeling.

To address nonlinear patterns, [3] employed Support Vector Regression and ensemble methods, demonstrating improved performance in capturing complex

relationships. Nevertheless, their work primarily emphasizes prediction accuracy without analyzing the effect of feature correlation.

Similarly, [2] applied machine learning techniques to real-world housing datasets, but lacks a comprehensive comparison between linear and nonlinear models under different evaluation strategies.

In contrast to these studies, our work provides a systematic comparison between linear models (OLS, Ridge, Lasso) and a nonlinear model (SVR), while also analyzing the impact of multicollinearity and evaluating model performance using both train-test split and cross-validation.

3 Methodology

3.1 Problem Definition

Given a real estate dataset with n samples, let $\mathbf{X} \in R^{n \times d}$ denote the feature matrix and $\mathbf{y} \in R^n$ denote the target variable representing house prices. Each sample includes attributes such as transaction date, house age, distance to the nearest MRT station, number of nearby convenience stores, and geographical location.

The objective of this study is to learn a function $f(\mathbf{X})$ that can accurately predict the house price $\mathbf{y}$. This is formulated as a supervised regression problem.

3.2 Data Preprocessing

Before training the models, several preprocessing steps are applied:

- **Handling missing values:** The dataset is checked for missing values and cleaned if necessary.
- **Feature selection:** Relevant features such as distance to MRT, number of stores, and location coordinates are retained.
- **Feature scaling:** All numerical features are normalized using standardization to ensure that models such as SVR are not biased toward features with larger scales.
- **Multicollinearity analysis:** In addition to the correlation matrix, the Variance Inflation Factor (VIF) is computed to quantitatively assess the degree of multicollinearity among features.

3.3 Models

We consider both linear and nonlinear regression models in this study.

OLS:

$$\hat{\beta} = \arg\min_{\beta} \|y - X\beta\|^2$$

where $X \in R^{n \times d}$ is the feature matrix, $y \in R^n$ is the target vector, and β is the coefficient vector.

Ridge:

$$\hat{\beta} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \|\beta\|_2^2$$

where $\lambda > 0$ is a regularization parameter controlling the penalty term.

Lasso:

$$\hat{\beta} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \|\beta\|_1$$

where $\|\beta\|_1 = \sum |\beta_j|$ promotes sparsity in the model coefficients.

SVR:

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad s.t. \quad |y_i - (w^T x_i + b)| \leq \epsilon$$

where ϵ defines the margin of tolerance and allows small prediction errors.

3.4 Evaluation Metrics

To evaluate model performance, we use the following metrics:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual values.
- **Root Mean Squared Error (RMSE):** Penalizes larger errors more heavily.
- R^2 **Score:** Indicates the proportion of variance explained by the model.

3.5 Training Strategy

To ensure a fair and robust evaluation, we adopt the following strategies:

- **K-fold Cross-Validation:** The dataset is divided into k folds, and each model is trained and validated k times to reduce variance in performance estimates.
- **Random Split:** The dataset is randomly divided into training and testing sets to provide a baseline evaluation.

4 Experiment

4.1 Dataset

The dataset used in this study is the Real Estate Valuation dataset, which contains information on property transactions collected in Taipei, Taiwan. The dataset consists of 414 samples and 6 input features, along with one target variable.

Each sample represents a real estate transaction and includes the following attributes:

- **X1: Transaction date** – the date when the property was sold, represented in a continuous numerical format.
- **X2: House age** – the age of the house in years.
- **X3: Distance to the nearest MRT station** – the distance (in meters) from the property to the nearest Mass Rapid Transit (MRT) station.
- **X4: Number of convenience stores** – the number of nearby convenience stores, representing accessibility to amenities.
- **X5: Latitude** – the geographical latitude of the property.
- **X6: Longitude** – the geographical longitude of the property.

The target variable is:

- **Y: House price of unit area** – the price per unit area of the property.

This dataset captures both spatial and non-spatial factors that influence real estate prices. In particular, location-related features such as distance to MRT, latitude, and longitude play an important role in determining housing value.

Before training the models, all numerical features are standardized to ensure consistent scaling across different variables.

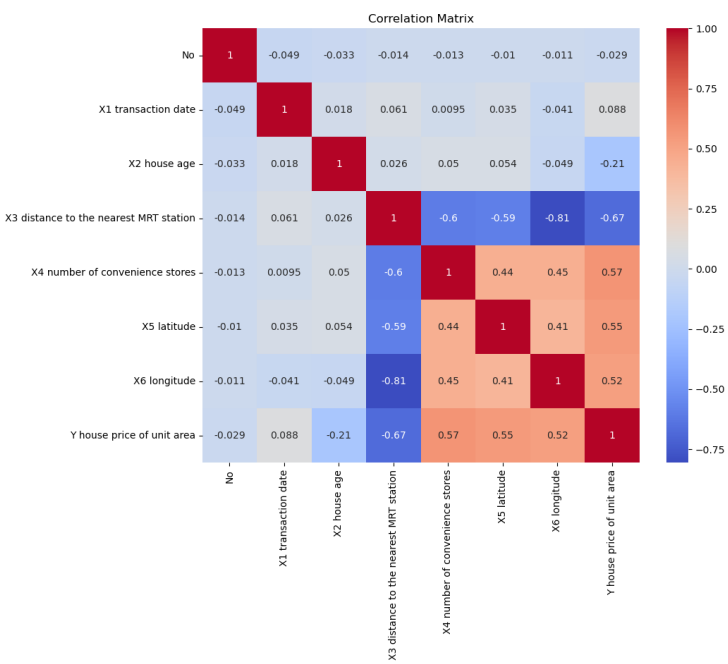

Figure 1: Correlation matrix of features

To further analyze multicollinearity, the Variance Inflation Factor (VIF) is computed for each feature. As shown in the results, most features have VIF values close to 1, indicating little to no correlation. The distance to the nearest MRT station has the highest VIF (approximately 4.33), suggesting a moderate level of correlation with other features. Overall, the results indicate that multicollinearity exists but is not severe in this dataset.

Feature	VIF
No	1.01
Transaction date	1.02
House age	1.02
Distance to MRT	4.33
Convenience stores	1.62
Latitude	1.61
Longitude	2.93

Table 1: Variance Inflation Factor (VIF) values

4.2 Results and Comparison

The performance of different models is summarized in Table X.

Model	MAE	RMSE	R^2
OLS	5.418033	7.389103	0.674541
Ridge	5.414440	7.384880	0.674913
Lasso	5.395214	7.347307	0.678213
SVR	5.786838	7.903483	0.627652

Table 2: Model performance comparison

In addition, the cross-validation results are shown in Table 3.

Model	CV RMSE
OLS	8.8040
Ridge	8.8028
Lasso	8.7875
SVR	8.9926

Table 3: Cross-validation results

5 Discussion

From the experimental results, several important observations can be made.

First, Lasso regression achieves the best overall performance, with the lowest RMSE (7.3473) and highest R^2 score (0.6782). While Lasso is known for

handling multicollinearity through feature selection, the VIF analysis indicates that multicollinearity in this dataset is not severe. Therefore, the performance improvement is more likely due to the regularization effect, which helps reduce overfitting and improve model stability.

Second, Ridge regression performs slightly better than OLS, suggesting that regularization contributes to improved model robustness. However, the relatively small improvement indicates that the dataset does not suffer from strong multicollinearity, which is consistent with the VIF results.

In contrast, SVR shows worse performance compared to linear models, with higher RMSE (7.9035) and lower R^2 (0.6277). This suggests that nonlinear relationships in the dataset are not strong enough for SVR to provide significant benefits.

The cross-validation results further confirm these findings. Lasso achieves the lowest CV RMSE (8.7875), indicating better generalization performance, while SVR has the highest CV RMSE, reinforcing that it is less suitable for this dataset.

Overall, the results suggest that the dataset is largely linear and does not exhibit severe multicollinearity. As a result, simple linear models are sufficient to capture the underlying relationships, and the use of more complex nonlinear models does not necessarily lead to improved performance.

6 Conclusion and Future Work

In this study, we compared linear and nonlinear regression models for real estate price prediction. The results show that regularized linear models, particularly Lasso, achieve the best performance, while nonlinear approaches such as SVR do not provide significant improvement for this dataset.

Further analysis using the Variance Inflation Factor (VIF) indicates that multicollinearity is not severe, suggesting that the dataset is relatively well-behaved. In addition, the experimental results imply that the relationships between features and house prices are largely linear. Therefore, simple linear models are sufficient to achieve strong performance in this problem.

For future work, more advanced models such as Gradient Boosting can be explored. Additionally, incorporating richer features, such as economic indicators or temporal trends, may further improve prediction accuracy.

References

[1] Li Chen and Jun Wang. A study on real estate price prediction using regression models. *Journal of Housing Research*, 2019.

[2] Soo Kim and Jin Park. Machine learning approaches for real estate valuation. *Applied Artificial Intelligence*, 2021.

[3] Wei Zhang and Ming Liu. Support vector regression for housing price prediction. *International Journal of Machine Learning*, 2020.